



NORTH SYDNEY BOYS HIGH SCHOOL

MATHEMATICS (EXTENSION 1)

2016 HSC Course Assessment Task 4

Thursday August 18, 2016

General instructions

- Working time – 50 minutes.
(plus 5 minutes reading time)
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the booklets used in the correct order within this paper and hand to examination supervisors.

SECTION I

- Mark your answers on the answer grid provided (on page 2)

SECTION II

- Commence each new question on a new page. Write on both sides of the paper.
- All necessary working should be shown in every question. Marks may be deducted for illegible or incomplete working.

STUDENT NUMBER: **# BOOKLETS USED:**

Class (please ✓)

☐ 12M3A – Mr Berry

☐ 12M4A – Mr Lin

☐ 12M3B – Ms Ziaziaris

☐ 12M4B – Dr Jomaa

☐ 12M3C – Miss Lee

☐ 12M4C – Mr Ireland

Marker's use only.

QUESTION	1-5	6	7	8	Total	%
MARKS	$\overline{5}$	$\overline{11}$	$\overline{11}$	$\overline{6}$	$\overline{33}$	

Answer sheet for Section I

Mark answers to Section I by fully blackening the correct circle, e.g “●”

STUDENT NUMBER:

Class (please ✓)

☐ 12M2A – Mr Berry

☐ 12M4A – Mr Lin

☐ 12M3B – Ms Ziaziaris

☐ 12M4B – Dr Jomaa

☐ 12M3C – Miss Lee

☐ 12M4C – Mr Ireland

1 – ☐ A ☐ B ☐ C ☐ D

2 – ☐ A ☐ B ☐ C ☐ D

3 – ☐ A ☐ B ☐ C ☐ D

4 – ☐ A ☐ B ☐ C ☐ D

5 – ☐ A ☐ B ☐ C ☐ D

Section I

5 marks

Attempt Question 1 to 5

Mark your answers on the answer grid provided.

Questions

Marks

1. The correct expression for $\binom{5}{k-1}$ is 1
- (A) $\frac{5!}{(k-1)!(4-k)!}$ (C) $\frac{5!}{(k-1)!(6-k)!}$
- (B) $\frac{5!}{(4-k)!}$ (D) $\frac{5!}{(6-k)!}$
2. A particle moving in simple harmonic motion with displacement x and velocity v is given by $v^2 = 9(16 - x^2)$. What is its amplitude (A) and its period (T)? 1
- (A) $A = 3, T = \frac{\pi}{2}$ (C) $A = 4, T = \frac{\pi}{2}$
- (B) $A = 3, T = \frac{2\pi}{3}$ (D) $A = 4, T = \frac{2\pi}{3}$
3. What is the coefficient of x^3 in the expansion of $(1 - 3x)^7$? 1
- (A) -945 (C) 35
- (B) -35 (D) 945
4. Which of the following is an equation for simple harmonic motion? 1
- (A) $x = a \cos nt$ (C) $x = a \cos nt - b \sin nt$
- (B) $x = a \sin(nt - \alpha)$ (D) All of the above
5. Which of the following is the coefficient of the term in x^n in the expansion of $(1+x)^n + (1-x)^n$? 1
- (A) 0 (C) $1 + (-1)^n$
- (B) $\frac{1 + (-1)^n}{2}$ (D) 2

Examination continues overleaf...

Section II

28 marks

Attempt Questions 6 to 8

Write your answers in the writing booklets supplied. Additional writing booklets are available. Your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (11 Marks)

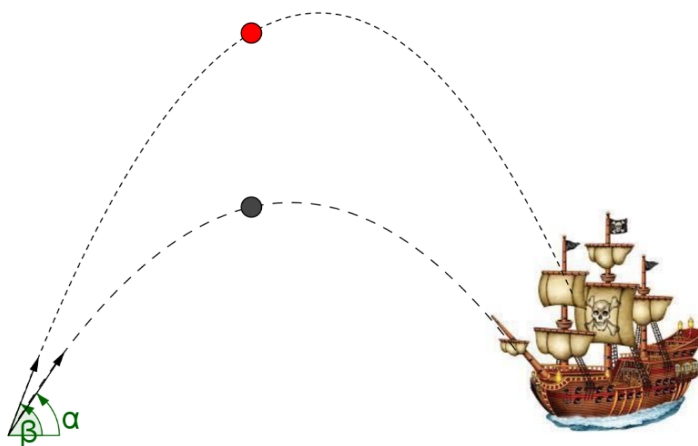
Commence a NEW page.

Marks

- (a) The velocity $v \text{ ms}^{-1}$ of a particle moving in simple harmonic motion along the x -axis is given by $v^2 = 8 - 2x - x^2$, where x is in metres.
- Find the acceleration of the particle in term of x . 1
 - State the centre and period of the motion. 2
 - What is the maximum speed of the particle? 1
- (b) A pirate ship is moored 560 m from a fort defending the harbor entrance of an island (see figure). The harbor defence cannon, located at sea level, has a muzzle velocity of 82 m/s. Assume that the position of the projectile is given by

$$x = vt \cos \theta, \quad y = vt \sin \theta - \frac{1}{2}gt^2,$$

where t is the time, in seconds, after firing, and g the acceleration due to gravity is 9.8 ms^{-2} .



- Show that the angles of elevation of the cannon to hit the pirate ship are 27° and 63° correct to the nearest degrees. 3
- What are the times of flight for the two elevation angles calculated above? 2
- How far should the pirate ship be from the fort if it is to be beyond range of the cannon? (Hint: Maximum range corresponds to an elevation angle of 45° .) 2

Please Turn Over

Question 7 (11 Marks)

Commence a NEW page.

Marks

- (a) A particle moves in a straight line and its position at time
- t
- is given by

$$x = 3 - \sqrt{3} \cos 2t + \sin 2t$$

- i. Prove that the particle is moving in simple harmonic motion about $x = 3$. **3**
 - ii. What is the period of the motion? **1**
 - iii. Find the amplitude of the motion. **2**
 - iv. When does the particle first reach maximum speed after time $t = 0$? **2**
- (b) Find the exact term independent of x in the expansion of **3**

$$(1 + 2x)^2 \left(2x + \frac{1}{x} \right)^{12}$$

Question 8 (6 Marks)

Commence a NEW page.

Marks

- (a) A rocket is fired from a pontoon on the sea. The rocket is aimed at a cliff which is
- h
- metres high and
- d
- metres from the pontoon. The angle of projection of the rocket is
- α
- and its initial velocity is
- $v \text{ ms}^{-1}$
- .

Take the point of projection as the origin O. Assume the acceleration due to gravity is 10 ms^{-2} . The horizontal component x and vertical component y of the position of the rocket at time t seconds are

$$x = vt \cos \alpha, \quad y = vt \sin \alpha - 5t^2.$$

(You are NOT required to derive these.)

- i. Show that the path of the rocket is given by the equation
- 1**

$$y = x \tan \alpha - \frac{5}{v^2} (1 + \tan^2 \alpha) x^2.$$

For questions (ii) and (iii), let $d = 240$, $h = 60$, $v = 40\sqrt{2}$, and $\alpha = 45^\circ$.

- ii. Find the time taken for the rocket to land on top of the cliff. **2**
- iii. Find the speed of the rocket and the impact angle when it reaches this point. **3**

End of paper.

Task 4 - 2016 - Ext 1

MC:

1. $\binom{5}{k-1} = \frac{5!}{(k-1)!(k-(k-1))!} = \frac{5!}{(k-1)!(6-k)!}$ (C)

2. $v^2 = 9(16 - x^2)$
 $\ddot{x} = \frac{d}{dn} \left(\frac{1}{2} v^2 \right) = \frac{d}{dn} \left(\frac{9}{2} (16 - x^2) \right) = -9x = -n^2 x \quad \therefore n=3$

$T = \frac{2\pi}{3}$. Also $v^2 = n^2(A^2 - x^2)$, $n=3$ and $A=4$

the amplitude is 4 and the period is $\frac{2\pi}{3}$. (D)

3. $(1-3x)^7 = \sum_{k=0}^7 \binom{7}{k} (-3x)^k = \sum_{k=0}^7 \binom{7}{k} (-3)^k x^k$

$k=3$, the coefficient of x^3 is $-27 \binom{7}{3} = -945$ (A)

4.

(A) $x = a \cos nt$
 $\dot{x} = -an \sin nt$
 $\ddot{x} = -an^2 \cos nt$
 $= -n^2 a \cos nt$
 $= -n^2 x$

(B) $x = a \sin(nt - \alpha)$
 $\dot{x} = an \cos(nt - \alpha)$
 $\ddot{x} = -n^2 a \sin(nt - \alpha)$
 $= -n^2 x$

(C) $x = a \cos nt - b \sin nt$
 $\dot{x} = -an \sin nt - bn \cos nt$
 $\ddot{x} = -an^2 \cos nt + bn^2 \sin nt$
 $= -n^2(a \cos nt - b \sin nt)$
 $= -n^2 x$

All the above

(D)

5. $(1+x)^n + (1-x)^n = \sum_{k=0}^n \binom{n}{k} x^k + \sum_{j=0}^n \binom{n}{j} (-x)^j$

the term in x^n , ($k=n$ and $j=n$) is

$\binom{n}{n} + \binom{n}{n}(-1)^n = 1 + (-1)^n$ (E)

question 6: (a)

$$(i) \quad v^2 = 8 - 2x - x^2 \quad \therefore \quad \frac{1}{2}v^2 = 4 - x - \frac{1}{2}x^2$$
$$\therefore \quad \frac{d(\frac{1}{2}v^2)}{dx} = -1 - \frac{1}{2} \times 2x = -(1+x)$$

$$(ii) \quad \text{centre of the motion is } x = -1 \text{ and } T = \frac{2\pi}{1} = 2\pi$$

$$(iii) \quad \ddot{x} = 0 \therefore x = -1$$

$$\therefore v^2 = 8 - 2(-1) - (-1)^2 = 9 \therefore v = 3.$$

(b)

$$(i) \quad x = v t \cos \theta, \quad y = v t \sin \theta - \frac{1}{2} g t^2$$
$$t = \frac{x}{v \cos \theta} \quad y = x \frac{\sin \theta}{\cos \theta} - \frac{1}{2} g \left(\frac{x}{v \cos \theta} \right)^2$$
$$y = x \frac{\sin \theta}{\cos \theta} - \frac{1}{2} g \frac{x^2}{v^2 \cos^2 \theta}$$

$$\frac{dy}{dx} = 0 \therefore \frac{\sin \theta}{\cos \theta} - g \frac{x}{v^2 \cos^2 \theta} = 0 \therefore \boxed{x = \frac{v^2 \sin \theta \cos \theta}{g}}$$

$$\text{Range} = \frac{2v^2 \sin \theta \cos \theta}{g} = \frac{v^2 \sin 2\theta}{g}$$

$$560 = \frac{82^2 \sin 2\theta}{9.8} \therefore \sin 2\theta = \frac{9.8 \times 560}{82^2} = 0.816$$

$$2\theta \approx 54^\circ \text{ or } 180 - 54^\circ$$

$$\therefore \boxed{\theta \approx 27^\circ \text{ or } 63^\circ}$$

(ii)

$$t_1 = \frac{x}{v \cos \theta} = \frac{560}{82 \cos 27} = 7.7 \text{ seconds}$$

$$t_2 = \frac{x}{v \cos \theta} = \frac{560}{82 \cos 63} = 15 \text{ seconds.}$$

$$(iii) \quad \theta = 45^\circ \therefore \text{Range} = \frac{v^2 \sin 90^\circ}{g} = \frac{v^2}{g} = \frac{82^2}{9.8} = 690 \text{ m}$$

Beyond that point the ship is safe.

Question 7:

(a) $x = 3 - \sqrt{3} \cos 2t + \sin 2t$

(i) $\dot{x} = -\sqrt{3}(-2 \sin 2t) + 2 \cos 2t$
 $= 2\sqrt{3} \sin 2t + 2 \cos 2t$

$\ddot{x} = 4\sqrt{3} \cos 2t - 4 \sin 2t$
 $= -4(-\sqrt{3} \cos 2t + \sin 2t)$
 $= -4(x - 3)$

Hence the result.

(ii) $T = \frac{2\pi}{n} = \frac{2\pi}{2} = \pi$

(iii) $A = \sqrt{(-\sqrt{3})^2 + 1^2} = \sqrt{4} = 2$

(iv) Maximum speed occurs at the centre of the motion ($x = 3$).

$\sin 2t = \sqrt{3} \cos 2t \therefore \tan 2t = \sqrt{3}$

$2t = \pi/3 \text{ and } t = \pi/6$

(b) $(1+2x)^2 \left(2x + \frac{1}{x}\right)^{12}$

$= (1+4x+4x^2) \sum_{k=0}^{12} \binom{12}{k} (2x)^{12-k} x^{-k}$

$= \sum_{k=0}^{12} \binom{12}{k} 2^{12-k} x^{12-2k} + \sum_{k=0}^{12} 4 \binom{12}{k} 2^{12-k} x^{13-2k} + \sum_{k=0}^{12} 4 \binom{12}{k} 2^{12-k} x^{14-2k}$

The term independent of x in $\sum_{k=0}^{12} \binom{12}{k} 2^{12-k} x^{12-2k}$

corresponds to $12-2k=0$ or $k=6$, setting $k=6$

we obtain $\binom{12}{6} 2^6$.

There is no term independent of x in $\sum_{k=0}^{12} 4 \binom{12}{k} 2^{12-k} x^{13-2k}$

since $k = \frac{13}{2}$ is not an integer value

The term independent of x in $\sum_{k=0}^{12} 4 \binom{12}{k} 2^{12-k} x^{14-2k}$

corresponds to $14-2k=0$ or $k=7$, setting $k=7$

we obtain $\binom{12}{7} \times 4 \times 2^5 = \binom{12}{7} 2^7$

Hence the term independent of x in $(1+2x)^2 \left(2x + \frac{1}{x}\right)^{12}$

is $\binom{12}{6} 2^6 + \binom{12}{7} 2^7$.

Question 8:

(a) (i) $t = \frac{x}{v \cos \alpha}$

$$y = \frac{x}{v \cos \alpha} \cdot v \sin \alpha - \frac{5}{v^2 \cos^2 \alpha} x^2$$

$$= x \tan \alpha - \frac{5}{v^2} x^2 (1 + \tan^2 \alpha)$$

$$\text{since } \frac{1}{\cos^2 \alpha} = \sec^2 \alpha = 1 + \tan^2 \alpha.$$

(ii) $60 = 40 \sqrt{2} \sin 45^\circ t - 5t^2$

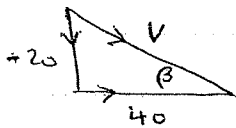
$$5t^2 - 40t + 60 = 0$$

$$t^2 - 8t + 12 = 0 \text{ or } (t-6)(t-2) = 0$$

$$\therefore t = 2 \text{ or } t = 6.$$

it takes 6 seconds to reach the top.

(iii)



$$\dot{x} = 40 \sqrt{2} \times \frac{1}{\sqrt{2}} = 40$$

$$\dot{y} = 40 \sqrt{2} \times \frac{1}{\sqrt{2}} - 10 \times 6 = -20$$

$$v^2 = (-20)^2 + 40^2 = 2000$$

$$v = 20\sqrt{5} \text{ m/s}$$

$$\tan \beta = \frac{20}{40} = \frac{1}{2} \therefore \beta = 26^\circ 34'$$
